

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2, BL6)

Assessment Tool Weightage: 30%

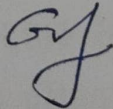
Annexure A

Industry Problem Solving Task

Code of Conduct:

I P. Ganesha (192424412) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed

at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost

implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimum cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with limited usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the

industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design

improves performance, enhances security, and allows expansion as the company grows.

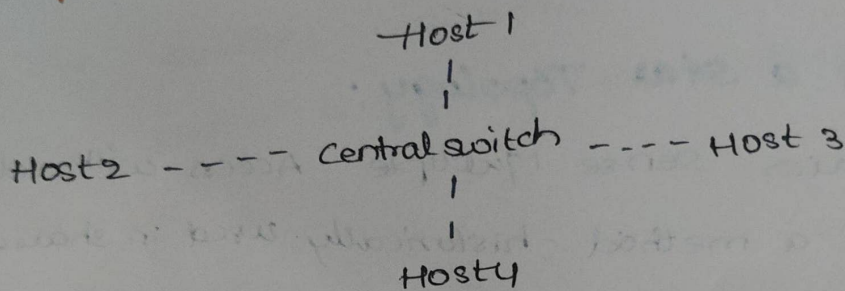
RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination	Good collaboration	Limited participation	No collaboration or contribution

Star topology Diagram and Description

A star topology is a network in which every host is connected individually to a central switch. The central switch acts as the main communication point between all connected devices.

Example :-



In this topology each host is connected to the switch using Cat 6 UTP cable. The switch device receives data from a source host and forwards it to the appropriate destination host.

For example a network with four hosts can have Host 1, Host 2, Host 3 and Host 4 connected to a central switch named SW1.

2. UTP segment length Validation

A UTP cable segment should be checked against the 100m maximum segment length specified in the assessment.

The validation can be explained as follows.

- If the cable length is less than or equal to 100m.
- If the cable length is greater than 100m.
- Negative values are invalid because a cable.

Test cases:-

Cable length	Result
50m	Valid
100m	Valid
120m	Invalid
150m	Invalid

Therefore, maintaining the cable length within the recommended length helps ensure reliable ethernet communication.

CSMA/CD in a star Topology.

CSMA/CD (Carrier sense Multiple Access with collision Detection) is a method historically used in shared ethernet networks to manage simultaneous transmission.

The basic operation is:-

1. A host first listens to the communication medium.
2. If the medium is free, the host starts transmission.
3. The hosts detect the collision.
4. The transmitting hosts stop transmitting.
5. Each host waits for a randomly selected back off interval.
6. After the back off period, the host attempts retransmission again.

For the four-host star network every collision event and its corresponding back off interval can be recorded in a log file.

Client Server Communication Through Star Switch

In a star network, two hosts can communicate through a central switch.

The communication process is:

Host 1 → switch → Host 2

The switch maintains a MAC Address table. This table associates MAC Addresses with switch ports.

For Example:-

MAC Address

AA:AA:AA:AA:AA:01

BB:BB:BB:BB:BB:02

connected Host

Host 1

Host 2

When Host 1 sends a frame to Host 2:-

1. Host 1 creates the Ethernet frame
2. The frame reaches the central switch
3. The switch examines the destination MAC address
4. The switch searches its MAC address table
5. The switch forwards the frame through that port
6. Host 2 receives the frame.

A client server program using the socket port can simulate this communication process and display the MAC.

5. Star Switch Operations:-

A star switch represents a central switch in a star topology.

It performs three important operations.

Add port :-

The add-port operation connects a new host to the switch. For example Host 1, Host 2, Host 3, and Host 4.

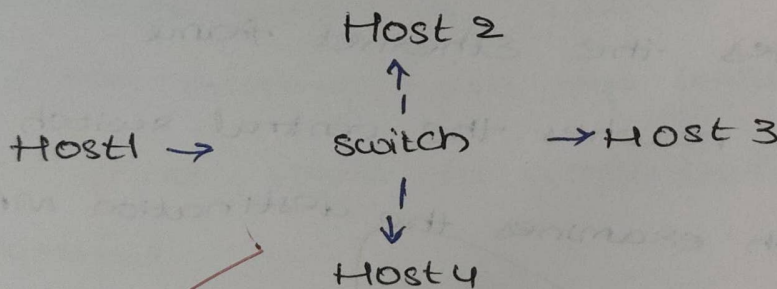
Remove port :-

The remove-port operation disconnects a host from the switch. Once removed, that host can be no longer.

Broad Cast :-

The broadcast operation sends a frame to all connected ports except the source port.

For Example



If Host 1 broadcast a frame Host 2 Host 3 Host 4 receive it, while Host 1 does not receive its own transmitted frame.